Gold dependency graphs: P2P sweep34 (33 problems)

Each problem's agent graph is built from the per-agent gold patches and file ownership, not from the conversations. Two agents are linked when one agent's gold slice uses a name the other's slice introduces or modifies (new def/class/CONSTANT, modified def, new keyword argument) or imports its module. Both directions and all shared names between a pair merge into ONE undirected link (counted once in E); arrows only show direction: A -> B means A's slice uses a name B's slice defines or changes. Links whose only shared name is a generic word are dropped. k = agents with at least one link, N = agents assigned, E = distinct linked pairs, density = 2E / [k(k-1)]. Star, chain and tree all have E = k-1, so their density is exactly 2/k.

topology	runs	rule	k range	density values seen	mean density	resolved
star	18	E = k-1, density = 2/k	2-6	1, 0.667, 0.5, 0.333	0.67	17/18
chain	3	E = k-1, density = 2/k	4-4	0.5	0.50	2/3
tree	2	E = k-1, density = 2/k	5-9	0.4, 0.222	0.31	2/2
mesh	7	E >= k, 2/(k-1) <= density <= 1	4-9	0.667, 0.619, 0.6, 0.5	0.58	5/7
disconnected	3	density varies	4-5	0.4, 0.333, 0.3	0.34	3/3
all	**33**		**2-9**		**0.58**	**29/33**

- Four stars have only 2 linked agents (a single link), so their density is 1.0 by construction.
- All kept links are drawn and counted as certain, including import-only, non-Python-endpoint and sibling-class links.
- The beliefs and no-interface arms run the same problems, so their graphs are identical to these.
- Known issue: prep.py drops a whole link when only ONE direction shares a generic word. This wrongly drops ansible cb94c0cc 2-3
- (shares add_tasknoplay_options; fixed: star 3/4, density 0.667) and openlibrary 6fdbbeee 3-6 (import of models.py; fixed: mesh 5/6,
- density 0.5). Graphs here follow the analysis as committed.
- Source: multiagent_pro_out_p2p_sweep34/analysis/comm_measures/prep/p2p/*.json; resolved from eval_out/eval_results.json.

gold link A -> B: A uses a name B defines/changes (thicker = more names)
agent with no gold link (in N only)
linked agent (in k), label = its gold file

Star (1/3)

18 problems | E = k-1, density = 2/k | one hub linked to every other linked agent; no spoke-spoke links

ansible 5640093f

resolved

k/N 2/4 E 1 density 1.000

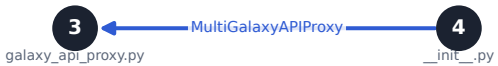


no link: 3 4

ansible a02e22e9

resolved

k/N 2/4 E 1 density 1.000

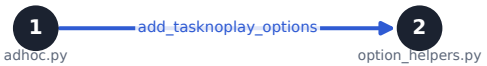


no link: 1 2

ansible cb94c0cc

resolved

k/N 2/4 E 1 density 1.000

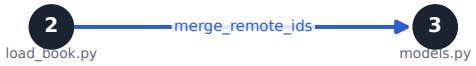


no link: 3 4

openlibrary 4b7ea297

unresolved

k/N 2/8 E 1 density 1.000

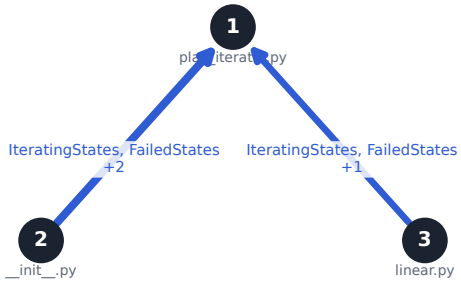


no link: 1 4 5 6 7 8

ansible 395e5e20

resolved

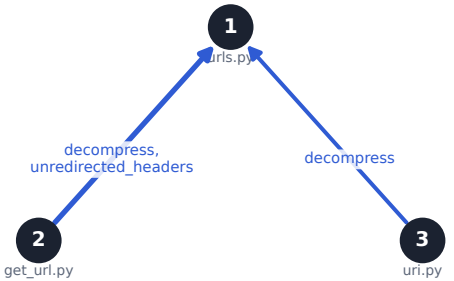
k/N 3/3 E 2 density 0.667



ansible d58e69c8

resolved

k/N 3/3 E 2 density 0.667



gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

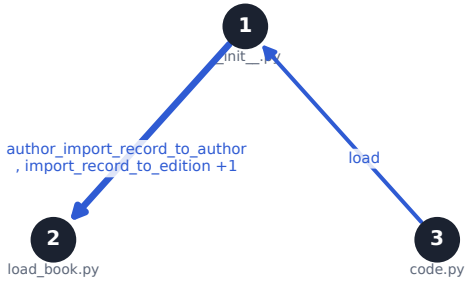
Star (2/3)

18 problems | E = k-1, density = 2/k | one hub linked to every other linked agent; no spoke-spoke links

openlibrary d40ec887

resolved

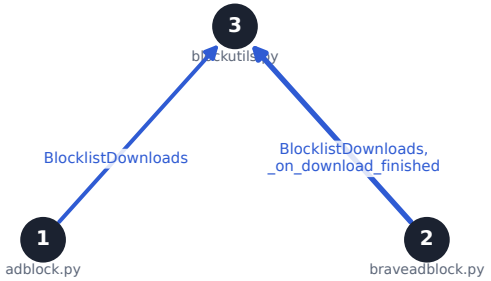
k/N 3/3 E 2 density 0.667



qutebrowser e57b6e0e

resolved

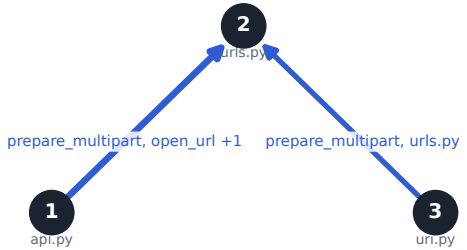
k/N 3/3 E 2 density 0.667



ansible b748edea

resolved

k/N 3/4 E 2 density 0.667

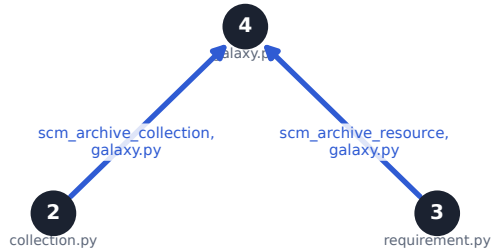


no link: 4

ansible e40889e7

resolved

k/N 3/4 E 2 density 0.667

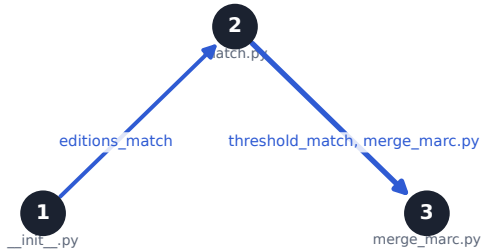


no link: 1

openlibrary 1be7de78

resolved

k/N 3/4 E 2 density 0.667

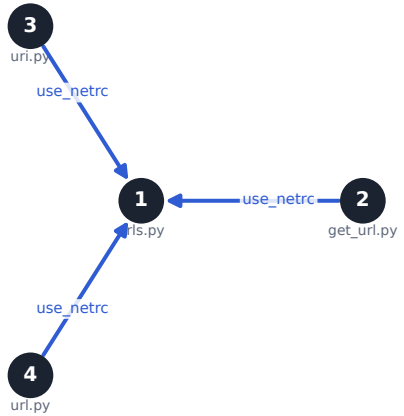


no link: 4

ansible a26c325b

resolved

k/N 4/4 E 3 density 0.500



gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

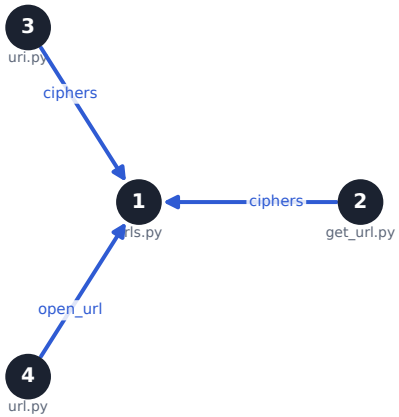
Star (3/3)

18 problems | E = k-1, density = 2/k | one hub linked to every other linked agent; no spoke-spoke links

ansible b8025ac1

resolved

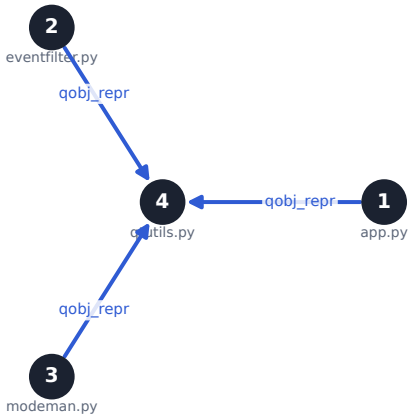
k/N 4/4 E 3 density 0.500



qutebrowser 0d2afd58

resolved

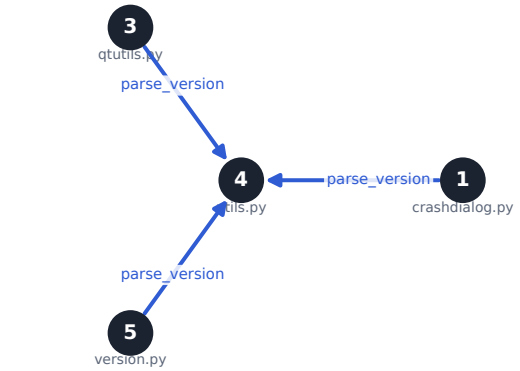
k/N 4/4 E 3 density 0.500



qutebrowser 46e6839e

resolved

k/N 4/5 E 3 density 0.500

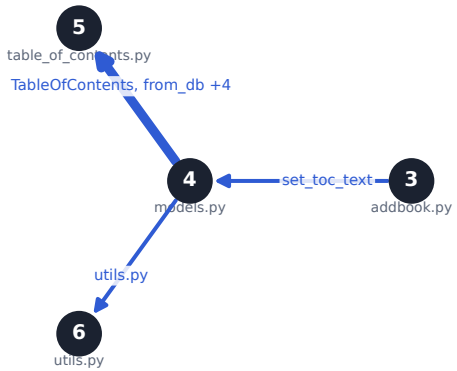


no link: 2

openlibrary 77c16d53

resolved

k/N 4/7 E 3 density 0.500

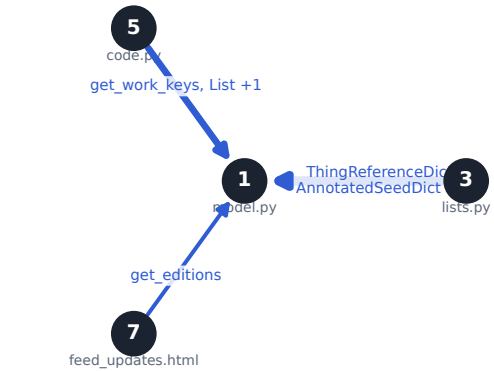


no link: 1 2 7

openlibrary d109cc7e

resolved

k/N 4/9 E 3 density 0.500

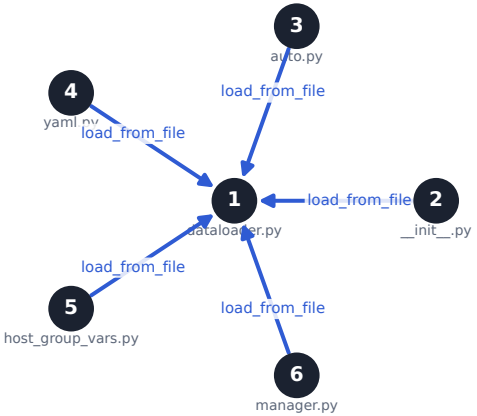


no link: 2 4 6 8 9

ansible 3b823d90

resolved

k/N 6/6 E 5 density 0.333



gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

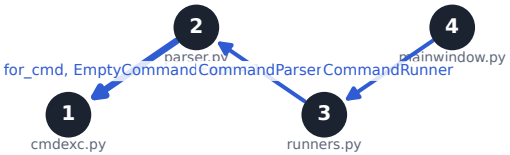
Chain

3 problems | E = k-1, density = 2/k | the linked agents form a single path

qutebrowser a84ecfb8

resolved

k/N 4/4 E 3 density 0.500



ansible 42355d18

unresolved

k/N 4/5 E 3 density 0.500



no link: 1

ansible 8127abbc

resolved

k/N 4/6 E 3 density 0.500



no link: 2 4

gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

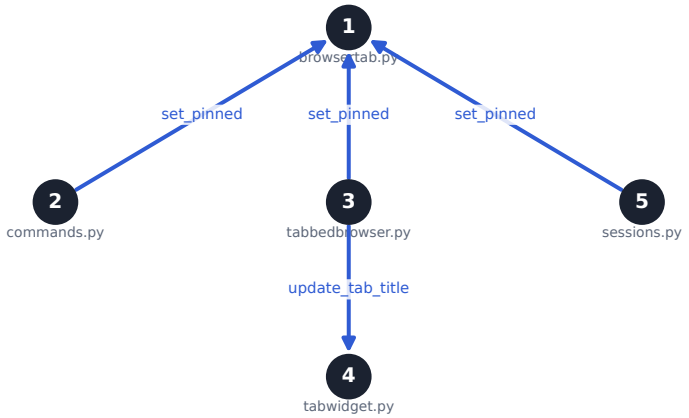
Tree

2 problems | E = k-1, density = 2/k | connected, no cycles, but not a path or star

qutebrowser 1943fa07

resolved

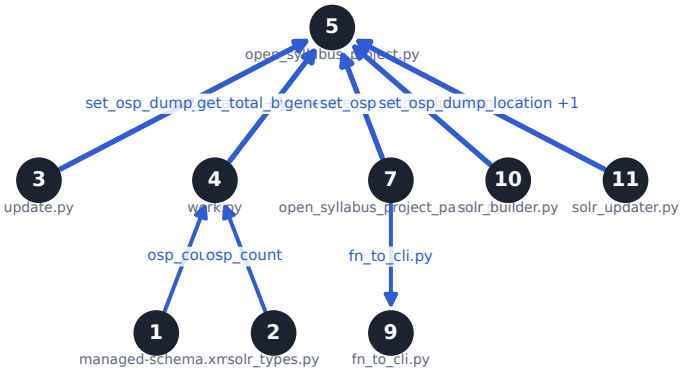
k/N 5/5 E 4 density 0.400



openlibrary c506c1b0

resolved

k/N 9/11 E 8 density 0.222



no link: 6 8

gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

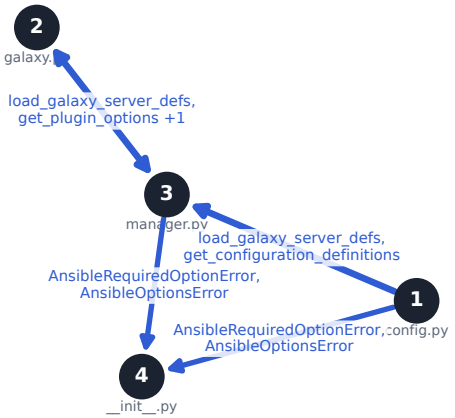
Mesh (1/2)

7 problems | E >= k, 2/(k-1) <= density <= 1 | connected and containing at least one cycle

ansible 949c503f

resolved

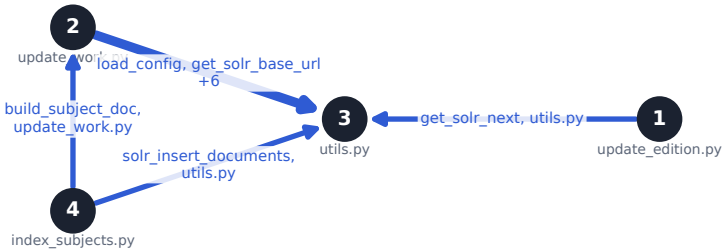
k/N 4/4 E 4 density 0.667



openlibrary 25858f9f

resolved

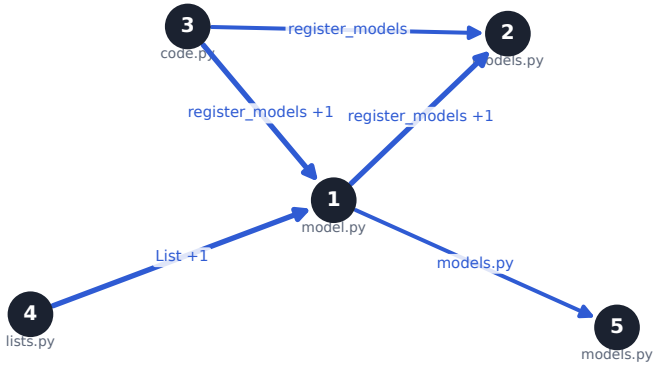
k/N 4/4 E 4 density 0.667



openlibrary 308a35d6

resolved

k/N 5/6 E 5 density 0.500

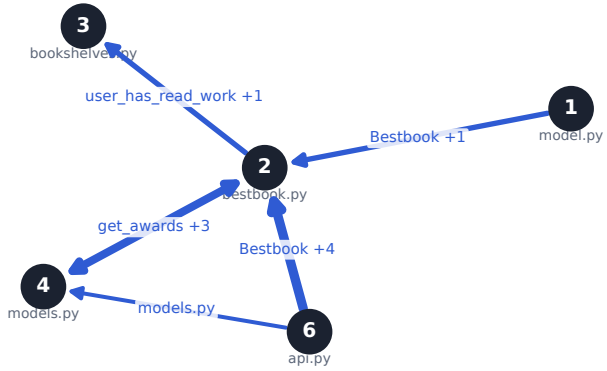


no link: 6

openlibrary 630221ab

unresolved

k/N 5/6 E 5 density 0.500



no link: 5

gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

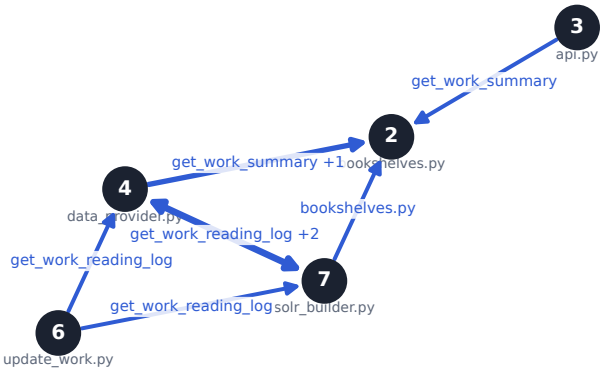
Mesh (2/2)

7 problems | E >= k, 2/(k-1) <= density <= 1 | connected and containing at least one cycle

openlibrary 7bf32385

unresolved

k/N 5/8 E 6 density 0.600

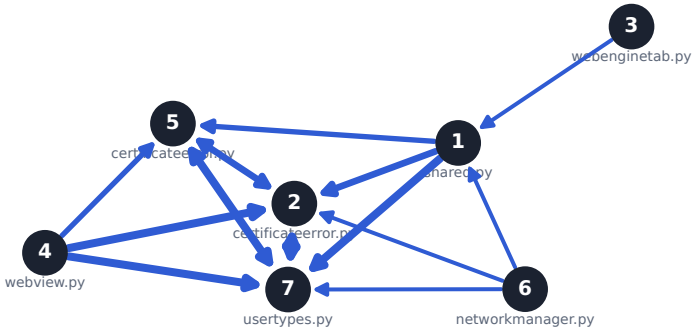


no link: 1 5 8

qutebrowser e5340c44

resolved

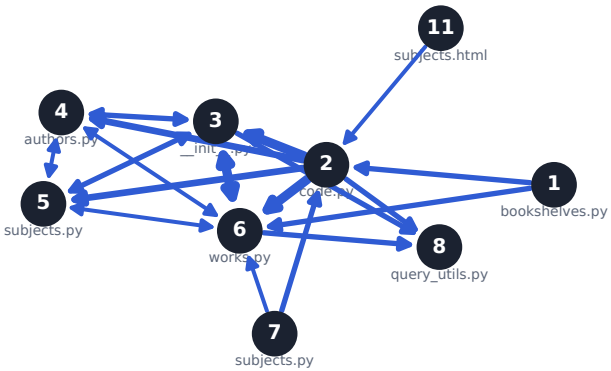
k/N 7/7 E 13 density 0.619 (shared names: see appendix)



openlibrary 7f6b722a

resolved

k/N 9/13 E 18 density 0.500 (shared names: see appendix)



no link: 9 10 12 13

gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

Disconnected

3 problems | density varies | the linked agents split into two or more separate groups

openlibrary 72321288

resolved

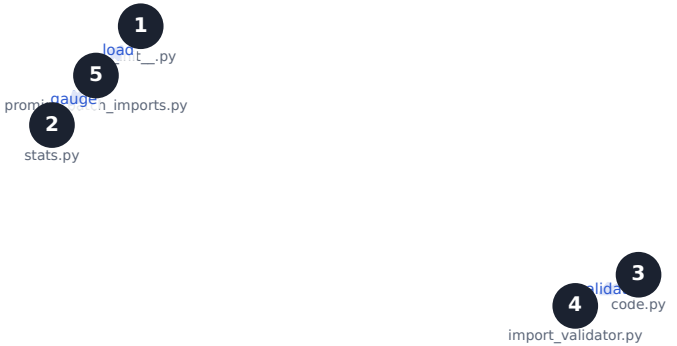
k/N 4/4 E 2 density 0.333



openlibrary b112069e

resolved

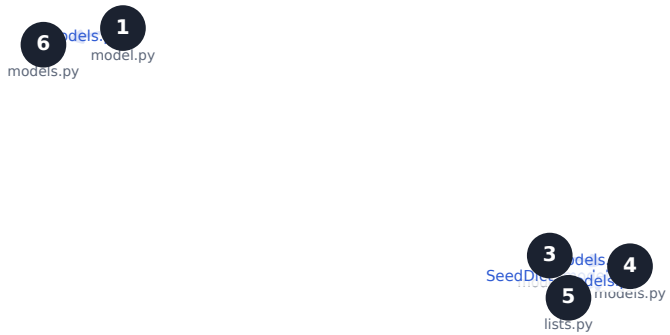
k/N 5/5 E 3 density 0.300



openlibrary 6fdbbeee

resolved

k/N 5/6 E 4 density 0.400



no link: 2

gold link A -> B: A uses a name B defines/changes (thicker = more names)

agent with no gold link (in N only)

linked agent (in k), label = its gold file

Appendix: links and shared names per problem

ansible 5640093f [star] k/N 2/4, E 1, density 1.000, resolved
agents: 1=module_common.py, 2=gather_facts.py, 3=package.py, 4=service.py
`1 <- 2: get_action_args_with_defaults`

ansible a02e22e9 [star] k/N 2/4, E 1, density 1.000, resolved
agents: 1=galaxy.py, 2=__init__.py, 3=galaxy_api_proxy.py, 4=__init__.py
`3 <- 4: MultiGalaxyAPIProxy`

ansible cb94c0cc [star] k/N 2/4, E 1, density 1.000, resolved
agents: 1=adhoc.py, 2=option_helpers.py, 3=console.py, 4=task_include.py
`1 -> 2: add_tasknoplay_options`
dropped (generic name only) 2-3: default, add_tasknoplay_options, add_runtask_options

openlibrary 4b7ea297 [star] k/N 2/8, E 1, density 1.000, UNRESOLVED
agents: 1=__init__.py, 2=load_book.py, 3=models.py, 4=import_edition_builder.py, 5=import_validator.py, 6=requirements.txt, 7=requirements_scripts.txt, 8=import_wikisource.py
`2 -> 3: merge_remote_ids`

ansible 395e5e20 [star] k/N 3/3, E 2, density 0.667, resolved
agents: 1=play_iterator.py, 2=__init__.py, 3=linear.py
`1 <- 2: IteratingStates, FailedStates, get_active_state, play_iterator.py`
`1 <- 3: IteratingStates, FailedStates, play_iterator.py`

ansible d58e69c8 [star] k/N 3/3, E 2, density 0.667, resolved
agents: 1=urls.py, 2=get_url.py, 3=uri.py
`1 <- 2: decompress, unredirected_headers`
`1 <- 3: decompress`

openlibrary d40ec887 [star] k/N 3/3, E 2, density 0.667, resolved
agents: 1=__init__.py, 2=load_book.py, 3=code.py
`1 -> 2: author_import_record_to_author, import_record_to_edition, build_query`
`1 <- 3: load`

qutebrowser e57b6e0e [star] k/N 3/3, E 2, density 0.667, resolved
agents: 1=adblock.py, 2=braveadblock.py, 3=blockutils.py
`1 -> 3: BlocklistDownloads`
`2 -> 3: BlocklistDownloads, _on_download_finished`

ansible b748edea [star] k/N 3/4, E 2, density 0.667, resolved
agents: 1=api.py, 2=urls.py, 3=uri.py, 4=uri.py
`1 -> 2: prepare_multipart, open_url, urls.py`
`2 <- 3: prepare_multipart, urls.py`

ansible e40889e7 [star] k/N 3/4, E 2, density 0.667, resolved
agents: 1=galaxy.py, 2=collection.py, 3=requirement.py, 4=galaxy.py
`2 -> 4: scm_archive_collection, galaxy.py`
`3 -> 4: scm_archive_resource, galaxy.py`

openlibrary 1be7de78 [star] k/N 3/4, E 2, density 0.667, resolved
agents: 1=__init__.py, 2=match.py, 3=merge_marc.py, 4=__init__.py
`1 -> 2: editions_match`
`2 -> 3: threshold_match, merge_marc.py`

ansible a26c325b [star] k/N 4/4, E 3, density 0.500, resolved
agents: 1=urls.py, 2=get_url.py, 3=uri.py, 4=url.py
`1 <- 2: use_netrc`
`1 <- 3: use_netrc`
`1 <- 4: use_netrc`

ansible b8025ac1 [star] k/N 4/4, E 3, density 0.500, resolved
agents: 1=urls.py, 2=get_url.py, 3=uri.py, 4=url.py
`1 <- 2: ciphers`
`1 <- 3: ciphers`

Appendix: links and shared names per problem

```
1 <- 4: open_url
```

qutebrowser 0d2afd58 [star] k/N 4/4, E 3, density 0.500, resolved
agents: 1=app.py, 2=eventfilter.py, 3=modeman.py, 4=qtutils.py
```
1 -> 4: qobj_repr
2 -> 4: qobj_repr
3 -> 4: qobj_repr
```

qutebrowser 46e6839e [star] k/N 4/5, E 3, density 0.500, resolved
agents: 1=crashdialog.py, 2=earlyinit.py, 3=qtutils.py, 4=utils.py, 5=version.py
```
1 -> 4: parse_version
3 -> 4: parse_version
4 <- 5: parse_version
```

openlibrary 77c16d53 [star] k/N 4/7, E 3, density 0.500, resolved
agents: 1=models.py, 2=TableOfContents.html, 3=addbook.py, 4=models.py, 5=table_of_contents.py, 6=utils.py, 7=view.html
```
3 -> 4: set_toc_text
4 -> 5: TableOfContents, from_db, to_db, from_markdown, to_markdown, table_of_contents.py
4 -> 6: utils.py
```

openlibrary d109cc7e [star] k/N 4/9, E 3, density 0.500, resolved
agents: 1=model.py, 2=SearchResultsWork.html, 3=lists.py, 4=utils.py, 5=code.py, 6=list.py, 7=feed_updates.html, 8=edit.html, 9=view_body.html
```
1 <- 3: ThingReferenceDict, AnnotatedSeedDict, get_editions, SeedDict, List, Seed, model.py
1 <- 5: get_work_keys, List, model.py
1 <- 7: get_editions
```
dropped (generic name only) 1-2: preview
dropped (generic name only) 2-6: name
dropped (generic name only) 2-7: with
dropped (generic name only) 1-8: value
dropped (generic name only) 6-8: name

ansible 3b823d90 [star] k/N 6/6, E 5, density 0.333, resolved
agents: 1=dataloader.py, 2=__init__.py, 3=auto.py, 4=yaml.py, 5=host_group_vars.py, 6=manager.py
```
1 <- 2: load_from_file
1 <- 3: load_from_file
1 <- 4: load_from_file
1 <- 5: load_from_file
1 <- 6: load_from_file
```

qutebrowser a84ecfb8 [chain] k/N 4/4, E 3, density 0.500, resolved
agents: 1=cmdexc.py, 2=parser.py, 3=runners.py, 4=mainwindow.py
```
1 <- 2: for_cmd, EmptyCommandError, NoSuchCommandError
2 <- 3: CommandParser
3 <- 4: CommandRunner
```

ansible 42355d18 [chain] k/N 4/5, E 3, density 0.500, UNRESOLVED
agents: 1=worker.py, 2=task_executor.py, 3=delegatable.py, 4=task.py, 5=manager.py
```
2 <-> 5: get_delegated_vars_and_hostname, TaskExecutor
2 <- 3: _calculate_delegate_to, TaskExecutor
4 <- 5: get_play
```

ansible 8127abbc [chain] k/N 4/6, E 3, density 0.500, resolved
agents: 1=worker.py, 2=task_executor.py, 3=task_queue_manager.py, 4=__init__.py, 5=loader.py, 6=__init__.py
```
1 -> 3: STDIN_FILENO, STDOUT_FILENO, STDERR_FILENO, task_queue_manager.py
1 -> 5: loader.py
5 <- 6: get_plugin_loader_namespace
```

qutebrowser 1943fa07 [tree] k/N 5/5, E 4, density 0.400, resolved
agents: 1=browsertab.py, 2=commands.py, 3=tabbedbrowser.py, 4=tabwidget.py, 5=sessions.py
```
1 <- 2: set_pinned
1 <- 3: set_pinned
3 -> 4: update_tab_title
```

Appendix: links and shared names per problem

```
1 <- 5: set_pinned
```

openlibrary c506c1b0 [tree] k/N 9/11, E 8, density 0.222, resolved
agents: 1=managed-schema.xml, 2=solr_types.py, 3=update.py, 4=work.py, 5=open_syllabus_project.py, 6=pyproject.toml, 7=open_syllabus_project_parser.py, 8=index-type.sh, 9=fn_to_cli.py, 10=solr_builder.py, 11=solr_updater.py

```
1 -> 4: osp_count
2 -> 4: osp_count
3 -> 5: set_osp_dump_location, open_syllabus_project.py
4 -> 5: get_total_by_olid, open_syllabus_project.py
5 <- 7: generate_osp_db, open_syllabus_project.py
7 -> 9: fn_to_cli.py
10 -> 5: set_osp_dump_location, open_syllabus_project.py
11 -> 5: set_osp_dump_location, open_syllabus_project.py
```

ansible 949c503f [mesh] k/N 4/4, E 4, density 0.667, resolved
agents: 1=config.py, 2=galaxy.py, 3=manager.py, 4=__init__.py

```
1 -> 3: load_galaxy_server_defs, get_configuration_definitions, get_config_value_and_origin
1 -> 4: AnsibleRequiredOptionError, AnsibleOptionsError
2 <-> 3: load_galaxy_server_defs, get_plugin_options, server_config_def
3 -> 4: AnsibleRequiredOptionError, AnsibleOptionsError
```

openlibrary 25858f9f [mesh] k/N 4/4, E 4, density 0.667, resolved
agents: 1=update_edition.py, 2=update_work.py, 3=utils.py, 4=index_subjects.py

```
1 -> 3: get_solr_next, utils.py
2 -> 3: load_config, get_solr_base_url, set_solr_base_url, get_solr_next, set_solr_next, SolrUpdateState, solr_update, utils.py
2 <- 4: build_subject_doc, update_work.py
3 <- 4: solr_insert_documents, utils.py
```

openlibrary 308a35d6 [mesh] k/N 5/6, E 5, density 0.500, resolved
agents: 1=model.py, 2=models.py, 3=code.py, 4=lists.py, 5=models.py, 6=utils.py

```
1 -> 2: register_models, models.py
1 -> 5: models.py
1 <- 3: register_models, model.py
2 <- 3: register_models
1 <- 4: List, model.py
```

openlibrary 630221ab [mesh] k/N 5/6, E 5, density 0.500, UNRESOLVED
agents: 1=model.py, 2=bestbook.py, 3=bookshelves.py, 4=models.py, 5=schema.sql, 6=api.py

```
1 -> 2: Bestbook, bestbook.py
2 -> 3: user_has_read_work, bookshelves.py
2 <-> 4: get_awards, Bestbook, get_count, bestbook.py
2 <- 6: Bestbook, AwardConditionsError, get_count, remove, bestbook.py
4 <- 6: models.py
```

openlibrary 7bf32385 [mesh] k/N 5/8, E 6, density 0.600, UNRESOLVED
agents: 1=managed-schema, 2=bookshelves.py, 3=api.py, 4=data_provider.py, 5=solr_types.py, 6=update_work.py, 7=solr_builder.py, 8=create-dump-table.sql

```
2 <- 3: get_work_summary
2 <- 4: get_work_summary, bookshelves.py
4 <-> 7: get_work_reading_log, WorkReadingLogSolrSummary, data_provider.py
4 <- 6: get_work_reading_log
6 -> 7: get_work_reading_log
2 <- 7: bookshelves.py
```

qutebrowser e5340c44 [mesh] k/N 7/7, E 13, density 0.619, resolved
agents: 1=shared.py, 2=certificateerror.py, 3=webenginetab.py, 4=webview.py, 5=certificateerror.py, 6=networkmanager.py, 7=usertypes.py

```
1 -> 2: defer, reject_certificate, accept_certificate
1 -> 5: defer, accept_certificate
1 -> 7: UndeferrableError, accept_certificate, reject_certificate, defer
2 <-> 5: defer, accept_certificate, reject_certificate
2 <-> 7: UndeferrableError, accept_certificate, reject_certificate, defer
1 <- 3: handle_certificate_error
2 <- 4: defer, reject_certificate, accept_certificate, create
```

Appendix: links and shared names per problem

```
4 -> 5: defer, accept_certificate
4 -> 7: accept_certificate, reject_certificate, defer, certificate_was_accepted
5 <-> 7: UndeferrableError, accept_certificate, reject_certificate, defer
1 <- 6: handle_certificate_error
2 <- 6: CertificateErrorWrapper
6 -> 7: certificate_was_accepted
```

openlibrary 7f6b722a [mesh] k/N 9/13, E 18, density 0.500, resolved

agents: 1=bookshelves.py, 2=code.py, 3=__init__.py, 4=authors.py, 5=subjects.py, 6=works.py, 7=subjects.py, 8=query_utils.py, 9=index.html, 10=authors.html, 11=subjects.html, 12=work_search.html, 13=__init__.py

```
1 -> 2: run_solr_query, code.py
1 -> 6: WorkSearchScheme, works.py
2 -> 3: SearchScheme, process_user_sort, process_user_query, build_q_from_params, q_to_solr_params
2 -> 4: AuthorSearchScheme, q_to_solr_params, authors.py
2 -> 5: SubjectSearchScheme, q_to_solr_params, subjects.py
2 -> 6: WorkSearchScheme, build_q_from_params, q_to_solr_params, has_solr_editions_enabled, works.py
2 -> 8: fully_escape_query, query_utils.py
3 <-> 4: q_to_solr_params, SearchScheme
3 <-> 5: q_to_solr_params, SearchScheme
3 <-> 6: WorkSearchScheme, is_search_field, transform_user_query, build_q_from_params, q_to_solr_params, works.py, SearchScheme
3 -> 8: fully_escape_query, query_utils.py
4 <-> 5: q_to_solr_params
4 <-> 6: q_to_solr_params
5 <-> 6: q_to_solr_params
6 -> 8: fully_escape_query, query_utils.py
2 <- 7: run_solr_query, code.py
6 <- 7: WorkSearchScheme
11 -> 2: get_results
```

openlibrary 72321288 [disconnected] k/N 4/4, E 2, density 0.333, resolved

agents: 1=works.py, 2=query_utils.py, 3=edition.py, 4=work.py

```
1 -> 2: luqum_replace_field
3 <- 4: translation_of
```

openlibrary b112069e [disconnected] k/N 5/5, E 3, density 0.300, resolved

agents: 1=__init__.py, 2=stats.py, 3=code.py, 4=import_validator.py, 5=promise_batch_imports.py

```
3 -> 4: validate
1 <- 5: load
2 <- 5: gauge
```

openlibrary 6fdbbeee [disconnected] k/N 5/6, E 4, density 0.400, resolved

agents: 1=model.py, 2=helpers.py, 3=model.py, 4=models.py, 5=lists.py, 6=models.py

```
1 -> 6: models.py
3 -> 4: models.py
3 <- 5: SeedDict, model.py
4 <- 5: models.py
```

dropped (generic name only) 3-6: models.py, type